

9.6.1 Expectation of Discrete random variable

Let X be a discrete random variable which takes the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$ then mathematical expectation of X denoted by $E(X)$ is defined by

$$E(X) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n.$$

$$= \sum_{i=1}^n x_i p_i \quad \text{where } \sum p_i = 1$$

$$E(X) = \sum_{i=1}^n x_i p(X = x_i)$$

Sometimes $E(X)$ is known as the mean of the random variable X .

Result:

If $g(X)$ is a function of the random variable X , then $E[g(X)] = \sum g(x)p(X=x)$

Properties:

- (i) $E(c) = c$ where c is a constant ✓
- (ii) $E(cX) = cE(X)$, where c is a constant
- (iii) $E(aX+b) = aE(X)+b$ ✓

Variance of Discrete random variable

Definition: In a probability distribution Variance is the average of sum of squares of deviations from the mean. The variance of the random variable X can be defined as.

$$\text{Var}(X) = E(X - E(X))^2$$

$$\boxed{\text{Var}(X) = E(X^2) - (E(X))^2}$$

Note Variance is the square of standard deviation.

Pb

When a die is thrown X denotes the number turns up. Find $E(X)$, $E(X^2)$ and $\text{Var}(X)$.

Soln Here X denote the number in the die

X	1	2	3	4	5	6
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \sum_{i=1}^6 x_i P(X=x_i)$$

$$= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

$$= \frac{21}{6}$$

$$E(X^2) = \sum_1^6 x_i^2 p(X=x_i)$$

$$[E(g(X)) = \sum_1^6 g(x) p(X=x)]$$

$$= 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} + 4^2 \cdot \frac{1}{6} + 5^2 \cdot \frac{1}{6} + 6^2 \cdot \frac{1}{6}$$

$$= \frac{91}{6}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{91}{6} - \left[\frac{21}{6}\right]^2 = \frac{100}{36} = \frac{25}{9}$$

Pb. -

The mean and standard deviation of a random variable X are 5 and 4 respectively

Find $E(X^2)$

Soln

given $E(X) = 5$

Standard deviation $(\sigma) = 4$

$$\therefore \text{Var}(X) = \sigma^2 = 16$$

We know $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$16 = E(X^2) - 5^2$$

$$\boxed{E(X^2) = 41}$$

9.6.2 Expectation of a continuous random variable

Let X be a continuous random variable with probability density function $f(x)$ then the mathematical expectation of X is defined as

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx \quad \text{provided the integral exists}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

Results:

$$(1) \quad E(c) = c \quad \text{where } c \text{ is constant}$$

$$(2) \quad E(aX) = a E(X)$$

$$\text{Variance} = E(X^2) - [E(X)]^2$$

$$\text{Variance} = (\text{Standard deviation})^2$$

$$E(X) = \text{mean}$$

Pb. —

The p.d.f. of a continuous random variable X is given by

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

find its mean and variance

Soln

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^2 x f(x) dx \\ &= \int_0^2 x \cdot \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^2 x^2 f(x) dx \\ &= \int_0^2 x^2 \cdot \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^4}{4} \right]_0^2 = \frac{1}{2} \left[\frac{2^4}{4} \right] \\ &= 2 \end{aligned}$$

$$\text{Variance} = E(X^2) - [E(X)]^2 = 2 - \frac{16}{9} = \frac{2}{9}$$

Pb

Suppose X is a continuous random variable with the probability density function given

$$\text{by } f(x) = \begin{cases} kx, & 0 \leq x \leq 2; \\ 2k, & 2 \leq x \leq 4, \\ 6k - kx, & 4 \leq x \leq 6. \end{cases}$$

Find the value of k , cumulative distribution function, mean, variance and standard deviation of X .

Soln Given f is a pdf so, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (6k - kx) dx = 1$$
$$\Rightarrow k \left[\frac{x^2}{2} \right]_0^2 + 2k [x]_2^4 + 6k[x]_4^6 - k \left[\frac{x^2}{2} \right]_4^6 = 1$$
$$k = \frac{1}{8}$$

$$f(x) = \begin{cases} \frac{x}{8}, & 0 \leq x \leq 2 \\ \frac{1}{4}, & 2 \leq x \leq 4 \\ \frac{3}{4} - \frac{x}{8}, & 4 \leq x \leq 6 \end{cases}$$

c.d.f

$$\text{if } x < 0, \quad F(x) = P(X \leq x) = 0$$

$$\text{if } 0 \leq x \leq 2, \quad F(x) = P(X \leq x) = \int_0^x \frac{t}{8} dt = \frac{x^2}{16}$$

$$\begin{aligned} \text{if } 2 \leq x \leq 4, \quad F(x) = P(X \leq x) &= \int_0^2 \frac{t}{8} dt + \int_2^x \frac{1}{4} dt \\ &= \frac{x-1}{4} \end{aligned}$$

$$\begin{aligned} \text{if } 4 \leq x \leq 6 \text{ then, } F(x) = P(X \leq x) &= \int_0^2 \frac{t}{8} dt + \int_2^4 \frac{1}{4} dt + \int_4^x \left(\frac{3}{4} - \frac{t}{8} \right) dt \\ &= \frac{3}{4}(x-3) - \frac{1}{16}(x^2-4) \end{aligned}$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^2 x \cdot \frac{x}{8} dx + \int_2^4 x \cdot \frac{1}{4} dx + \int_4^6 x \left(\frac{3}{4} - \frac{x}{8} \right) dx$$

=

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^2 x^2 \frac{x}{8} dx + \int_2^4 x^2 \frac{1}{4} dx + \int_4^6 x^2 \left(\frac{3}{4} - \frac{x}{8} \right) dx$$

=

$$\text{Var}(X) = E(X^2) - [E(X)]^2 =$$

$$\text{Standard deviation} = \sqrt{\text{Var}(X)}$$

1. Statement for Discrete random variable

If X and Y are two discrete random variables then

$$E(X+Y) = E(X) + E(Y)$$

2. Statement for Continuous random variable

Let X and Y are two continuous random variables with probability density functions $f(x)$ and $f(y)$ respectively. Then $E(X+Y) = E(X) + E(Y)$

Discrete random variable:

Statement: If X and Y are two independent variables then $E(XY) = E(X) E(Y)$

Continuous random variable:

Statement: If X and Y are independent random variables Then $E(XY) = E(X) E(Y)$

$$E(X+Y) = E(X) + E(Y)$$

$$E(XY) = E(X) \cdot E(Y)$$

Some results based on variance:

- (i) $\text{Var}(c) = 0$ i.e. Variance of a constant is zero
- (ii) If c is constant then $\text{Var}(cX) = c^2 \text{Var}(X)$
- (iii) If X is a random variable and c is a constant then $\text{Var}(X \pm c) = \text{Var}(X)$
- (iv) a and b are constants then $\text{Var}(aX \pm b) = a^2 \text{Var}(X)$
- (v) a and b are constants then $\text{Var}(a \pm bX) = b^2 \text{Var}(X)$.
- (vi) If X and Y are independent random variables then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$